Duality Theorem:

- Duality is an important principle of Boolean algebra. Most of the postulates are combined in pairs.
- We can define them as part (i) and part (ii).
- If binary operators and identity elements are interchanged each other then it is possible to drive one part of postulate from another. This process is known as duality.

Possible changes are:

- Change each (+) sign to a (•) sign
- Change each (•) sign to a (+) sign
- Complementing each 0 to 1

Example:

We know from postulate 5(i)

$A + 0 = A$

If we want to drive the dual relation of this Boolean relation then we will find

$A \bullet 1 = A$

This is similar to postulate 5(ii).

Here (+) sign is changed by (•) sign and 0 is changed by its complement, i.e. 1.

DeMorgan's Theorem:

- The DeMorgan's theorem is another most useful mathematical theorem in Boolean algebra.
- It simplifies the Boolean expression in which a product or sum of elements is inverted.

The pairs are

- $\overline{A + B} = \overline{A} \cdot \overline{B}$
- $\overline{A \bullet B} = \overline{A} + \overline{B}$

Proof

The Demorgan's theorem also can be explained using different values of X and Y. For first theorem

X	Y	X+Y	(X+Y)'
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

X	Y	X'	Y'	X'.Y'
0	0	1	1	1
0	1	1	0	0
1	0	0	1	0
1	1	0	0	0

- It can be seen that for different values of x and y, both table show similar results.
- Therefore, we can say $\overline{(X+Y)} = X'Y'$

Proof

For second theorem

X	Y	X.Y	(X.Y)'
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

X	Y	X'	Y'	X'+Y'
0	0	1	1	1
0	1	1	0	1
1	0	0	1	1
1	1	0	0	0

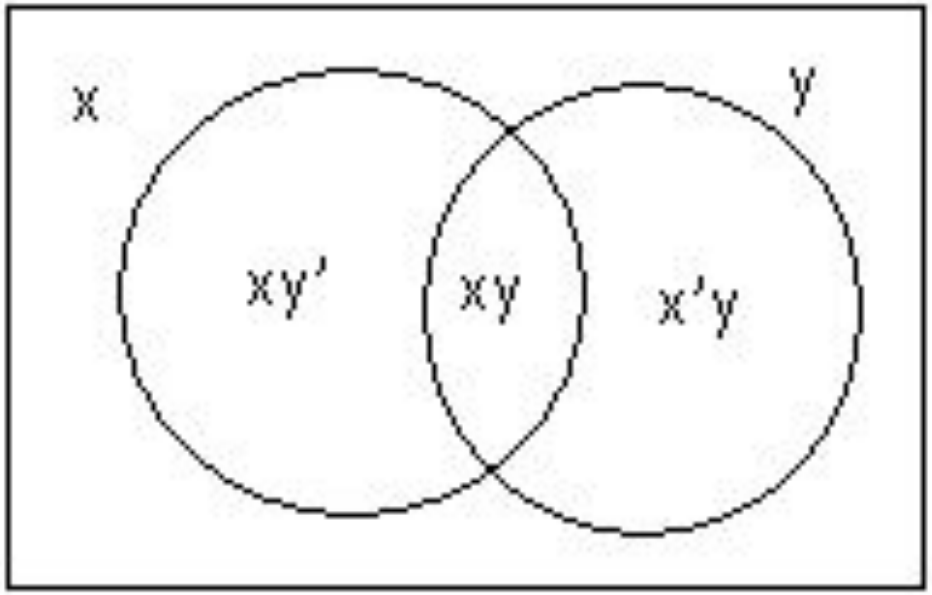
- From this table, we can say

$$\overline{X.Y} = \bar{X} + \bar{Y}$$

- The two tables are identical, and so the two expressions are identical.

Venn Diagram

The Venn diagram visualizes the relationship among the variables of a Boolean expression.



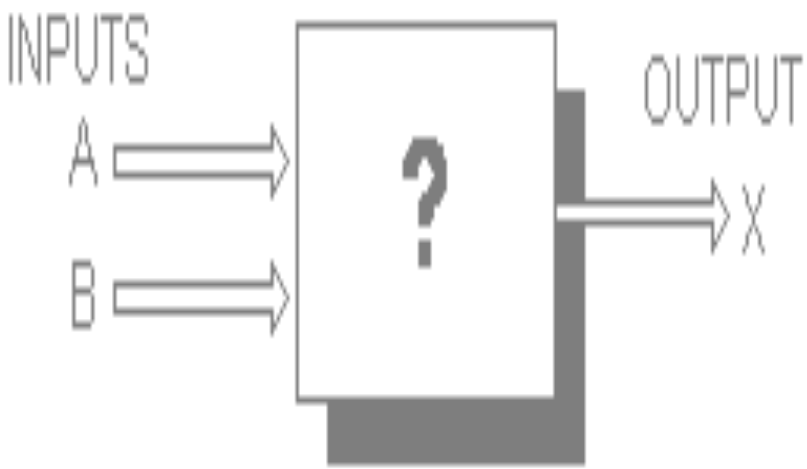
Boolean Functions

- **What is Boolean function?**
- A Boolean function is an expression formed with binary variables, two binary operators (AND, OR), unary operator NOT, parentheses and equal sign.
- **Suppose:**
- F= X+(X'Y+Z)
- It may also be represented in a truth table.
- Need 2^n combination of 1's or 0's

Truth Table

- A truth table is a means for describing how a logic circuit's output depends on the logic levels present at the circuit's inputs.
- Lets two variable X, Y and their output is F.

$2^n=2^2=4$



A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

- Table 2-2: Lets $n=3$, $2^n=2^3=8$

x	y	z	F1	F2	F3	F4
0	0	0	0	0	0	0
0	0	1	0	1	1	1
0	1	0	0	0	0	0
0	1	1	0	0	1	1
1	0	0	0	1	1	1
1	0	1	0	1	1	1
1	1	0	1	1	0	0
1	1	1	0	1	0	0